

Blockchain

Networked Systems

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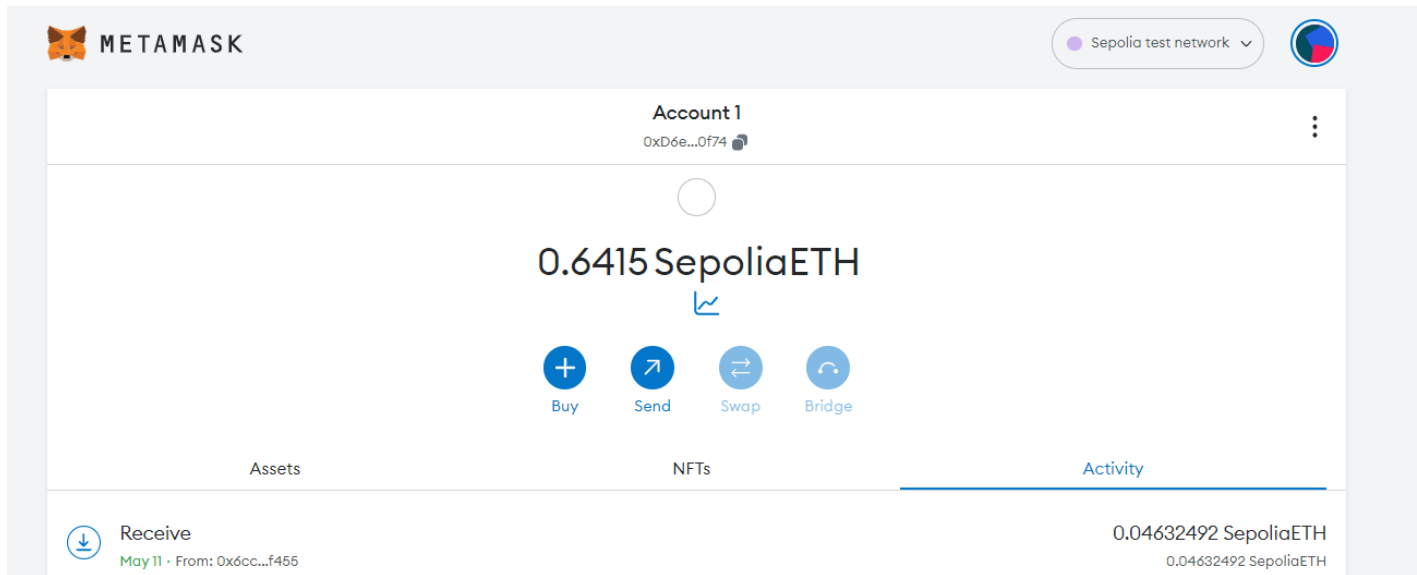
Spring 2026

Outline

- Logistic
- Basics Intro to Blockchain
- Solidity Basics
- Smart Contracts on Sepolia Test Network
- Account:
 - 0xD6eE2c7359e51f1330167A6B66afd515a0Bf0f74

Wallets

- Task 0
 - Create Ethereum wallet using Metamask
 - <https://metamask.io/>
 - Create wallet, and switch to Sepolia Testnet
 - Change view to include test networks



Wallets

- Task 1
 - Open Sepolia Faucet page
 - <https://sepolia-faucet.pk910.de/>

Wallets

- Task 1
 - Copy account id and check it on etherscan
 - <https://sepolia.etherscan.io/>
 - It is block explorer
- *ToDo Task 1*
 - Add screen of the etherscan web page with your account to the report

History and Evolution

- Lets start with the Internet history

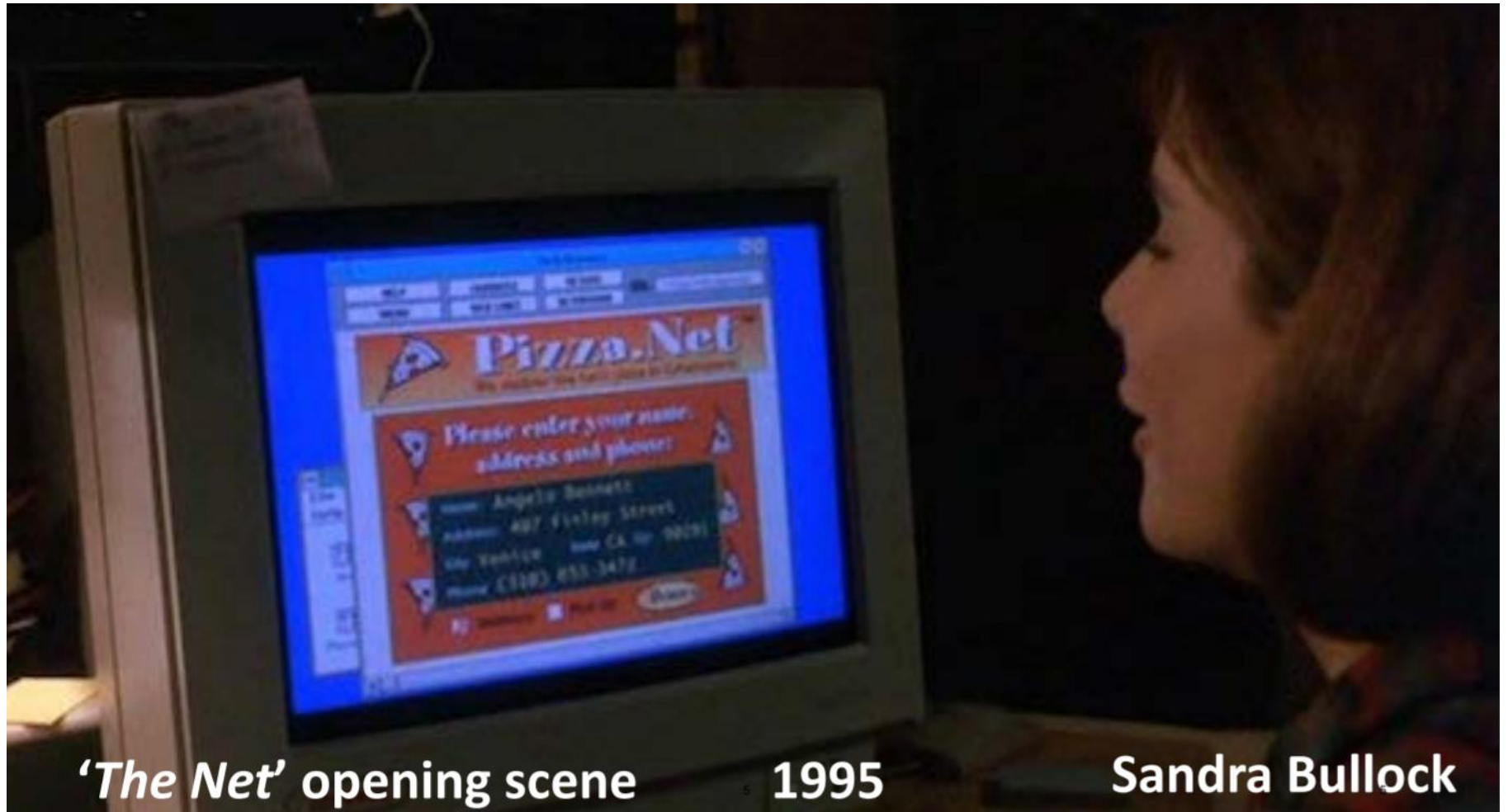
HTTP- 1990

TCP/IP - 1974

Ethernet - 1974



History and Evolution



'The Net' opening scene

1995

Sandra Bullock

History and Evolution

www.pizzahut.com/assets/pizza Search

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Document URL: http://www.pizzahut.com/



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Name :

Street Address:

Voice Phone ###-###-####
 (where we can reach you)



Cryptography

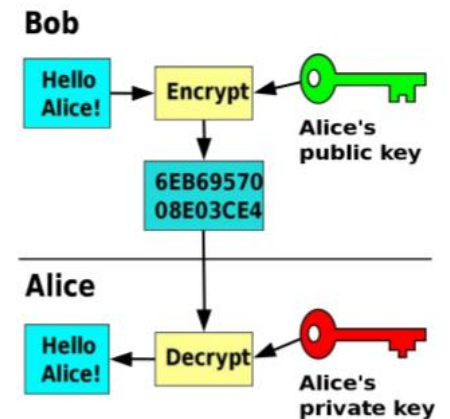
- Communication in the presence of adversaries



Scytale Cipher
Ancient Times



Enigma Machine
1920s - WWII



Asymmetric Cryptography
1976 to today

History: Cryptographic

- It is basics of modern internet

SSL / TLS – 1996



HTTP- 1990



TCP/IP - 1974



Ethernet - 1974



Early Cryptographic Digital Currencies

- DigiCash (David Chaum) – 1989
- Mondex (National Westminster Bank) - 1993
- CyberCash (Lynch, Melton, Crocker & Wilson) – 1994
- E-gold (Gold & Silver Reserve) – 1996
- Hashcash (Adam Back) – 1997
- Bit Gold (Nick Szabo) – 1998
- B-Money (Wei Dai) – 1998
- Lucre (Ben Laurie) – 1999

Digital Payments Innovation



Bitcoin: A Peer-to-Peer Electronic Cash System

- From: Satoshi Nakamoto vistomail.com>
Subject: Bitcoin P2P e-cash paper
Newsgroups: gmmane.com.comp.encryption.general
Date: Friday 31st October 2008 18:10:00 UTC
- “I've been working on a new electronic cash system that's fully peer-to-peer, with no trusted third party.”

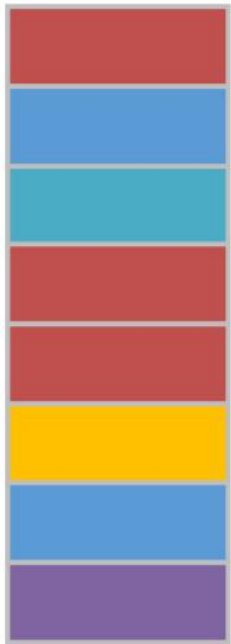
A new layer?

- Programmable transactions
- What is the future ?



What is a blockchain?

timestamped
append-only log



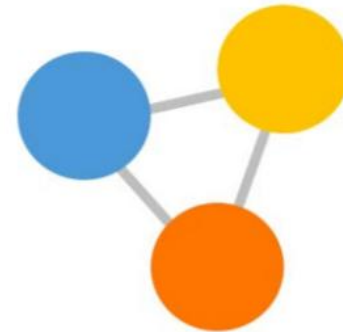
auditable database



Secured via cryptography

- Hash functions for **tamper resistance** and **integrity**
- Digital signatures for **consent**
- Consensus for **agreement**

consensus protocol



Addresses '**cost of trust**'
(Byzantine Generals problem)

- Permissioned
- Permissionless

Pizza for bitcoins ?

Bitcoin Forum

simple machines forum

April 10, 2021, 11:27:04 AM

Welcome, **Guest**. Please [login](#) or [register](#).

News: Latest Bitcoin Core release: [0.21.0](#) [Torrent]



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Author

Topic: Pizza for bitcoins? (Read 797853 times)

laszlo

Full Member



Activity: 199

Merit: 519



Pizza for bitcoins?

May 18, 2010, 12:35:20 AM

Merited by DaRude (50), Seccour (50), Vod (20), sapta (20), alani123 (12), LFC_Bitcoin (11), OgNasty (10), Nomad88 (10), Totscha (10), TimtheYoutuber (10), the_poet (10), arthurbonora (10), leps (10), mnightwaffle (10), suchmoon (9), cheefbuza (7), d5000 (5), Betwrong (5), bitbollo (5), ebliever (5), krogothmanhattan (5), LiteBit (5), mia_houston (5), nutildah (3), klondike_bar (3), vapourminer (2), BitcoinFX (2), LoyceV (2), gbianchi (2), cygan (2), bones261 (2), Halab (2), ChiBitCTy (2), fillippone (2), crypto_curious (2), ivaxmm (2), malevolent (1), EFS (1), JayJuanGee (1), iluvbitcoins (1), HI-TEC99 (1), UnDerDoG81 (1), batang_bitcoin (1), ETFbitcoin (1), S3cco (1), coolcoinz (1), digit (1), TheQuin (1), Astargath (1), lukax8 (1), jacktheking (1), frankenmint (1), bitart (1), Julien_Olympic (1), o_e_l_e_o (1), AlcoHoDL (1), JanEmil (1), amishmanish (1), apoorvlathey (1), DireWolfM14 (1), elianite (1), Toxic2040 (1), VB1001 (1), pushups44 (1), chimk (1), BobLawblaw (1), taserz (1), Financisto (1), invincible49 (1), nullius (1), GazetaBitcoin (1), tim-bc (1), fishfishfish313 (1), SimpleFX (1), thirdprize (1), BTC Liz (1), Toughit (1), barjan (1), M-BTC (1), dektex (1), lonchafina (1), grinbuck (1), alia (1), inkling (1), Kda2018 (1) #1

I'll pay 10,000 bitcoins for a couple of pizzas.. like maybe 2 large ones so I have some left over for the next day. I like having left over pizza to nibble on later. You can make the pizza yourself and bring it to my house or order it for me from a delivery place, but what I'm aiming for is getting food delivered in exchange for bitcoins where I don't have to order or prepare it myself, kind of like ordering a 'breakfast platter' at a hotel or something, they just bring you something to eat and you're happy!

I like things like onions, peppers, sausage, mushrooms, tomatoes, pepperoni, etc.. just standard stuff no weird fish topping or anything like that. I also like regular cheese pizzas which may be cheaper to prepare or otherwise acquire.

If you're interested please let me know and we can work out a deal.

Thanks,
Laszlo

BC: 157fRrqAKrDyGHR1Bx3yDxeMv8Rh45aUet

Pizza for bitcoins ?

- May 21, 2010, 07:06:58 PM
 - “So nobody wants to buy me pizza? Is the bitcoin amount I'm offering too low?”
- May 22, 2010, 07:17:26 PM
 - “I just want to report that I successfully traded 10,000 bitcoins for pizza. Pictures:
<http://heliacal.net/~solar/bitcoin/pizza/>
Thanks jercos!”

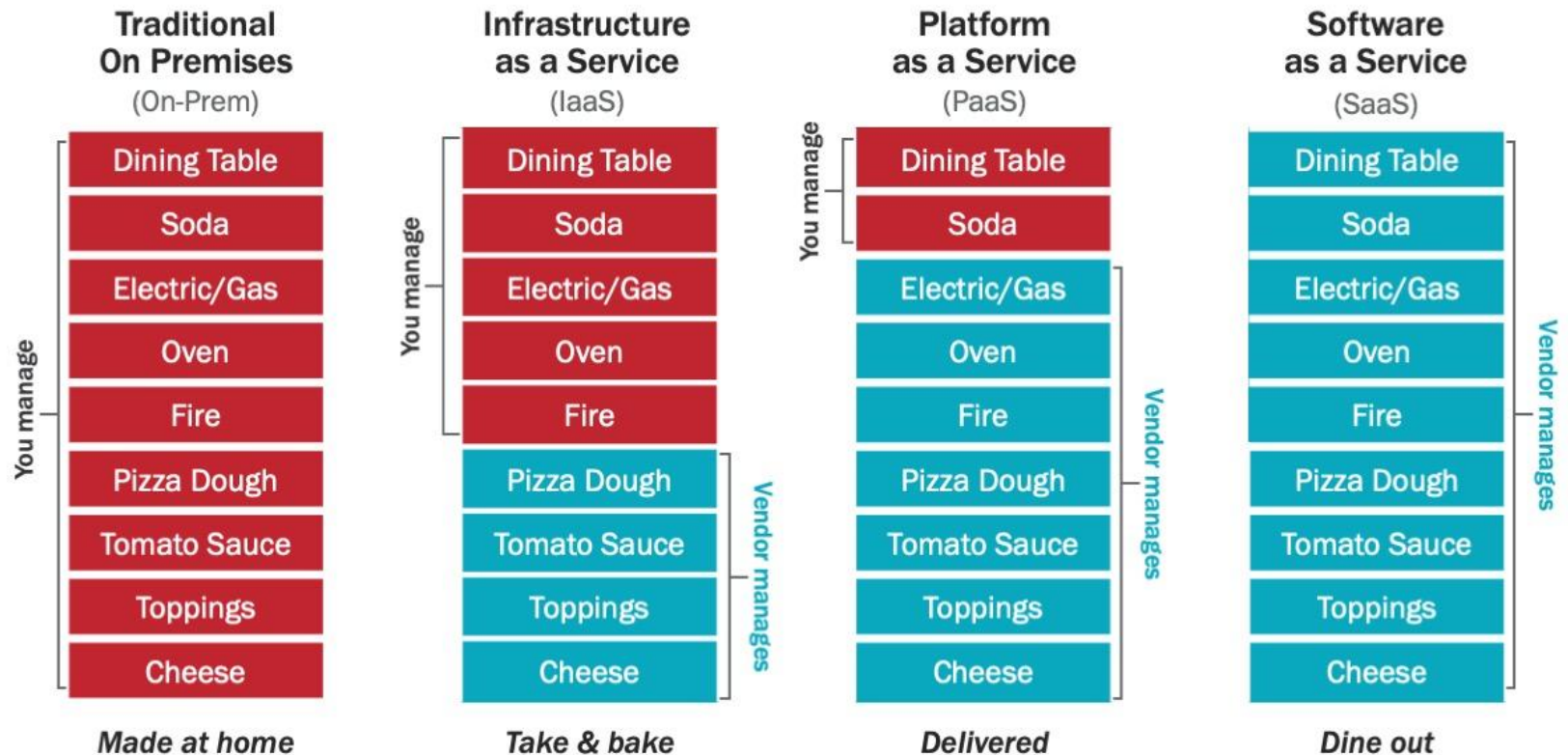


Pizza for bitcoins ?

- May 22, 2010 - \$41
 - \$20.50 per pizza
- April 2025 - \$ 766 233 600.00
 - 2 980 294 052.78 PLN
- May 22, Bitcoin Pizza Day

Pizza side notes

Pizza as a Service



Blockchain Intro

- 2008 – bitcoin paper
 - Bitcoin as a protocol over blockchain
 - <https://andersbrownworth.com/blockchain/>
- 2015 – Ethereum paper and product
 - Smart contract agreements without intermediaries
 - Smart contracts are written in code and executed in decentralized network
- That is main difference between bitcoin and Ethereum
 - It uses Merkle Patricia-Trie
 - It uses Proof of Stake

Bitcoin protocol

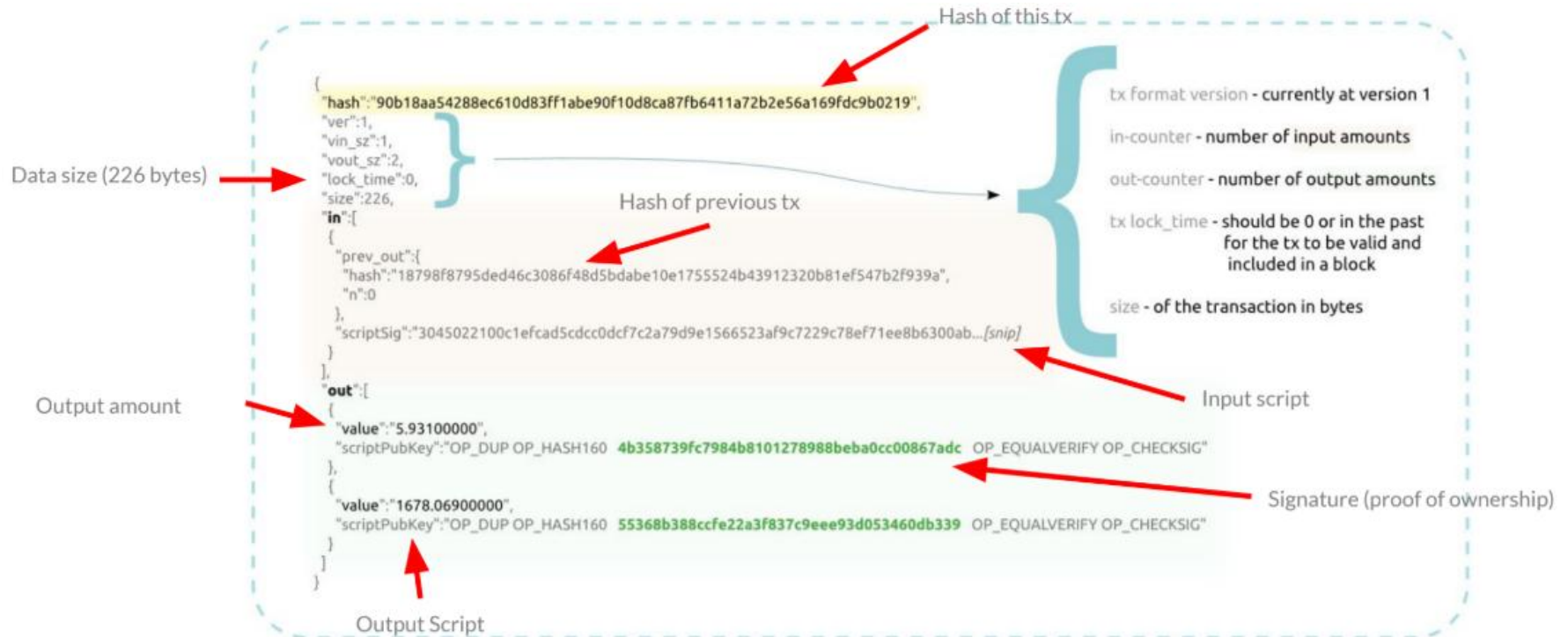
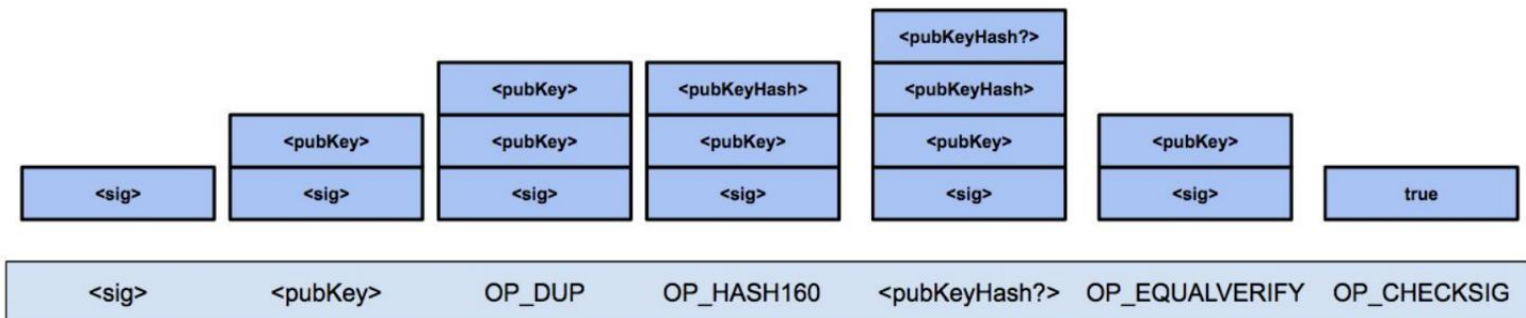


image by Vervan <vervan@gmail.com> 2014 CC SA



DApp

- DApp == Smart Contracts == Decentralized Applications
- DAOs
 - Decentralized Autonomous Organization
- Web3 Idea

Web Evolution

- Web 1
 - Starting in 1990, dot com era
 - Similarly, like tv and radio => read web
- Web 2
 - Read-write web, though technology development
 - Social media, but also mobile web with geo-awareness
 - It democratize publishing

Web Evolution

- Web 3
 - Creators have difficulties with monetization
 - Read-write-own web
 - We own digital assets we create
 - Users become owners
 - Blockchain networks store tokens
 - Everything you create can be represented as a token
- More recent idea with the Solid Project
 - <https://solidproject.org/>

Agentic Payments

- Coinbase Agentic Wallets
 - Software limits transactions and spending, non-custodian architecture
 - Over 50 million transactions for the X402 protocol

The screenshot shows the Coinbase Developer Platform website. The navigation bar includes links for Cryptocurrencies, Individuals, Businesses, Institutions, Developers, and Company. There are also search, help, and sign-in/sign-up buttons. The main heading is "Introducing Agentic Wallets: Give Your Agents the Power of Autonomy", dated February 11, 2026. Below the heading is a promotional graphic with the text "Give your agent a wallet" and terminal-style commands: "# Give your agent a wallet", "\$ npx skills add coinbase/agent-wallet-skills", "# or try it yourself", and "\$ npx awal --help". To the right of the graphic is a "Create account" section with the text "It's free and takes 2 minutes or less." and a "Get started" button with a right arrow.

Coinbase Developer Platform

Developer Platform > Discover > Launches

Introducing Agentic Wallets: Give Your Agents the Power of Autonomy

February 11, 2026

Give your agent a wallet

```
# Give your agent a wallet
$ npx skills add coinbase/agent-wallet-skills

# or try it yourself
$ npx awal --help
```

coinbase DEVELOPER PLATFORM

Create account
It's free and takes 2 minutes or less.

Get started →

Features

- Decentralized
 - Run by node operators in a distributed
 - Compare it to centralized banks
- Transparency and flexibility
 - Pseudo anonymous
- Speed and efficiency
 - For making a transaction
- Security and immutability
 - We are not able to change the data
- Removal of counterparty risk
 - e.g., insurance agreement (who makes profits)
- Trust minimized agreements
 - Trustless (based on math rather than brands)

Wallets

- Gas: Measure of computing use
- Gas Price: How much it costs per unit of gas
- Gas Limit: Max amount of gas in transaction
- Transaction fee: gas used x gas price
 - <https://eth-converter.com/>
- Mining is process of finding solution to blockchain problem
 - Computing hash for blocks

Network

- Nodes builds the network
 - Decentralized
 - Anyone can join the network
 - Blockchain is resilient
- Consensus
 - Mechanisms/protocol used to agree on a state
- Two characteristics:
 - Chain selection => Nakamoto consensus
 - Sybil resistance => PoW and PoS
 - Layer 1 and Layer 2 networks

Remix IDE

- We will use Remix IDE
 - <https://remix.ethereum.org/>
- Basic types:
 - <https://docs.soliditylang.org/en/v0.8.26/types.html>
- Basic contracts and language features
 - <https://docs.soliditylang.org/en/v0.8.26/contracts.html>

Remix IDE

- DEMO

Remix IDE

- Functions have to be specified as being external, public, internal or private.
- State modification:
 - Writing to state variables.
 - Emitting events.
 - Creating other contracts.
 - Using *selfdestruct*.
 - Sending Ether via calls.
 - Calling any function not marked view or pure.
 - Using low-level calls.
 - Using inline assembly that contains certain opcodes.

Task 2.1

- Use RemixVM in this part of Task 2
- Create smart contract for accepting donation
 - Payable function
 - Use *msg.value*
- Keep track of the depositors
 - Map address and stored value (using mapping)

```
// adding entries to mapping  
identitiesToAmountFunded[msg.sender] += msg.value;
```

Task 2.1

- Once you collect all money, allow withdraw only for contract creator

- In constructor:

- // the first person to deploy the contract is the owner

- constructor() {

- owner = payable(msg.sender);

- // getting balance of the deployed contract

- address(this).balance;

Task 2.2

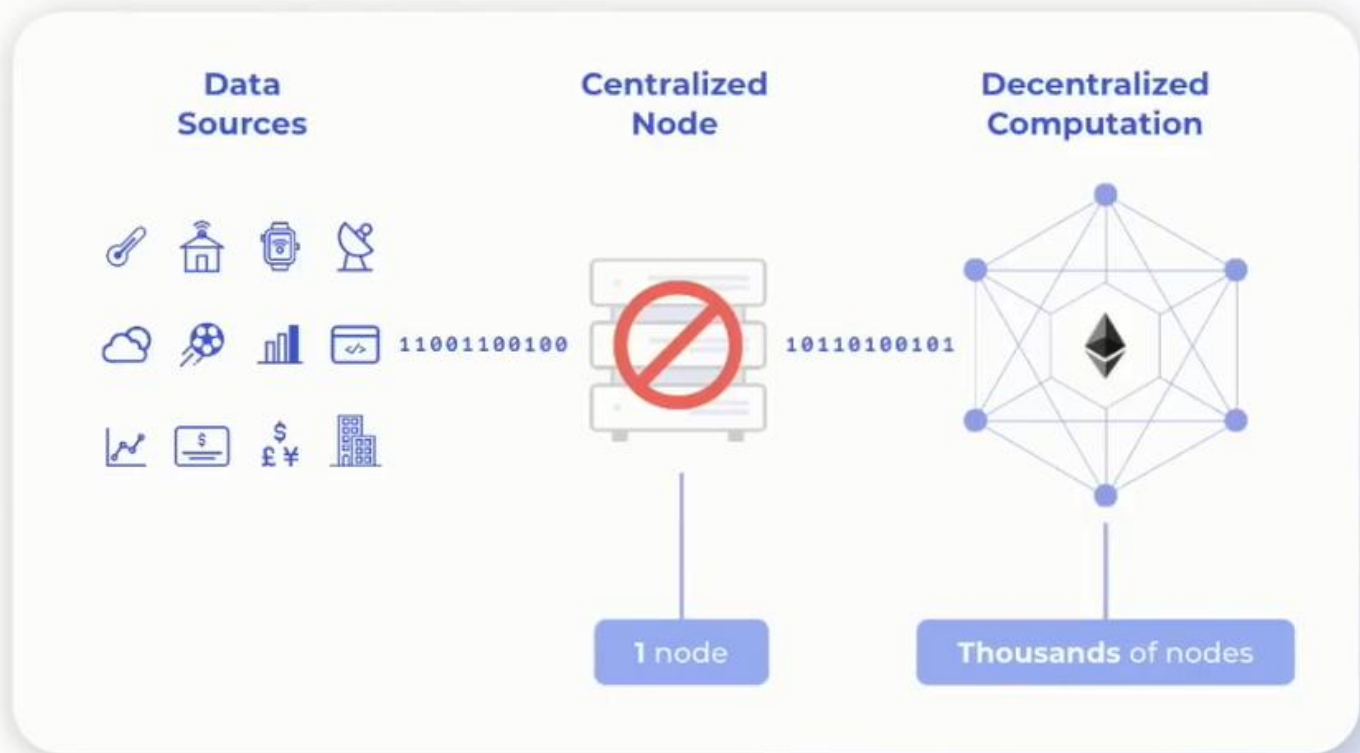
- Deploy on Sepolia Testnet
- Accept minimum 1 Euro donation
 - <https://docs.chain.link/data-feeds/using-data-feeds>
 - You need to convert Ether -> Euro, use 2 different interfaces
- *ToDo*
 - In report copy your source code of your contract
 - Add etherscan history of your account
 - MAKE ETHER TRANSFER

Blockchain Intro

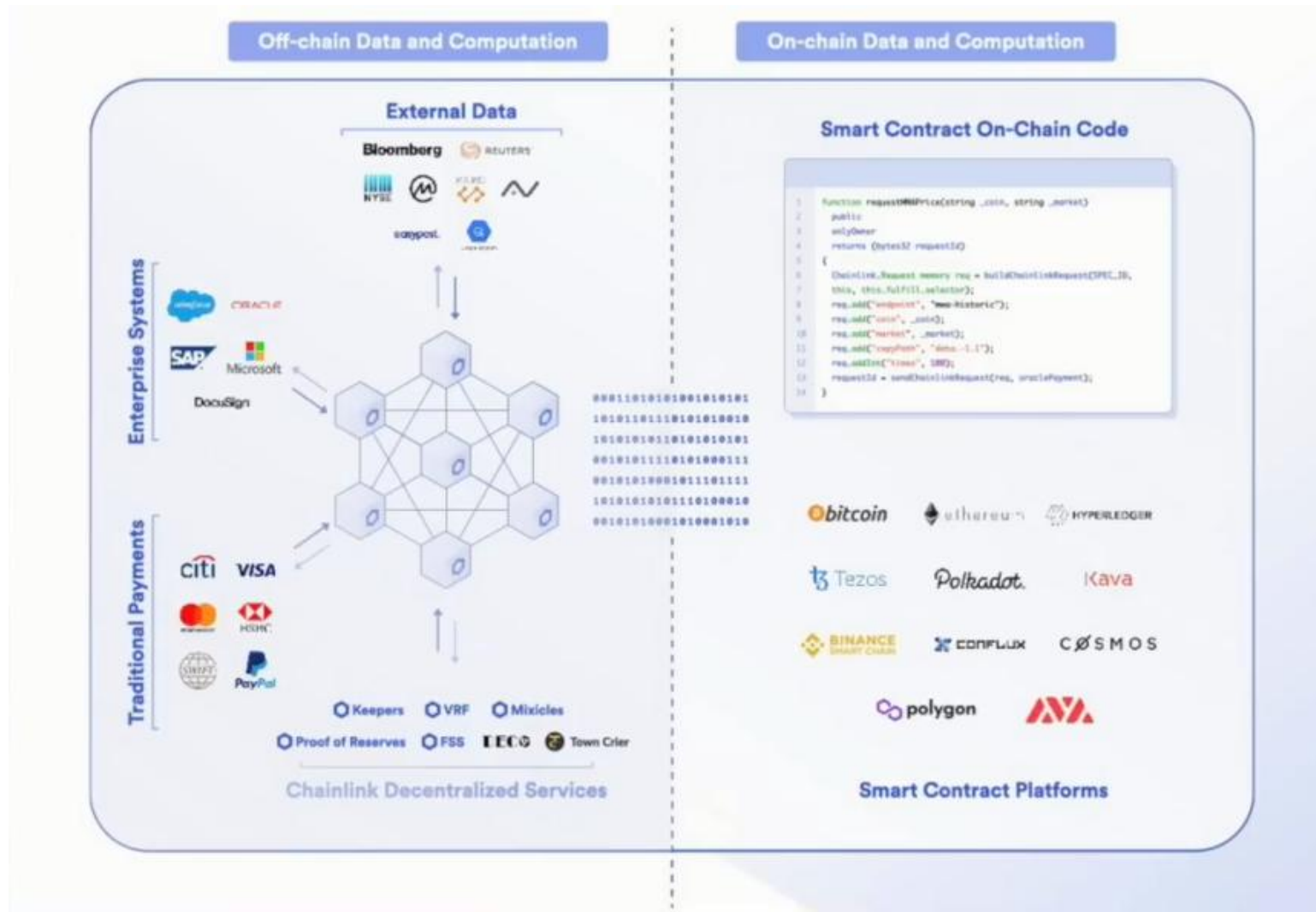
- Oracle problem
 - Smart contracts are executed in sandboxes
 - They need real world data
 - Oracles are entities that brings data into the chain

Blockchain Intro

Centralized Oracles are a Point of Failure



Hybrid Smart Contracts



Thank you